

Reproducing Classical Heuristics for Urban Rail Transit Line Planning

Laporte 2005 & Ahmed 2020

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Outline

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The Transit Network Design Problem (TNDP)

- Design an urban rail transit network: **where to build stations** and **how to connect them**.
- Combines station location, alignment planning, and network topology under constraints:
 - Construction budgets
 - Population/trip coverage
 - Geometric feasibility (non-intersection, spacing)
- NP-hard in the general case $\Rightarrow$ practical heuristics needed.

Our Goal

Reproduce two representative algorithms and evaluate them on a Shanghai dataset with 54 candidate stations.

Overview of the Two Papers

Paper 1: Laporte et al. 2005

“Maximizing Trip Coverage in the Location of a Single Rapid Transit Alignment”

- Single line
- Maximize OD coverage
- Greedy heuristics
- Polynomial-time

Paper 2: Ahmed et al. 2020

“GIS and Genetic Algorithm Based Integrated Optimization for Rail Transit System Planning”

- Multi-line network
- Minimize total system cost
- Genetic algorithm
- GIS integration

Candidate Station Selection (Shanghai)

- 54 candidate stations in central Shanghai
- Selected from 4 spatial data layers:
 - Population centers
 - Job centers
 - Transport hubs
 - Commercial centers
- Data source: OpenStreetMap
- Corridor graph built for alignment search



MTCP (Maximal Trip Coverage Problem):

- Build **one non-intersecting path** through candidate stations
- Maximize total OD trip coverage
- Subject to a **length budget** $L_{\max}$

$$\max \sum_{(i,j) \in OD} f_{ij} \cdot \mathbf{1}[\text{trip } (i,j) \text{ is covered}]$$

$$\text{s.t. total line length} \leq L_{\max}$$

- Generalizes bounded-length Hamiltonian path $\Rightarrow$ **NP-hard**
- f_{ij} : gravity-model OD demand between stations i and j

H1: Greedy Extension

- 1 Start from a seed edge
- 2 Extend at either endpoint
- 3 Pick the node with highest marginal coverage gain
- 4 Repeat until budget exhausted

Complexity: $O(n^2 \cdot |E|^2)$

H2: Greedy Insertion

- 1 Start from a seed edge
- 2 Insert new node at *any* position in path
- 3 Pick (node, position) with highest gain
- 4 Repeat until budget exhausted

Complexity: $O(n^2 \cdot |E|^3)$

Three variants: **H2-A** (max ridership), **H2-B** (min length), **H2-C** (max ratio)

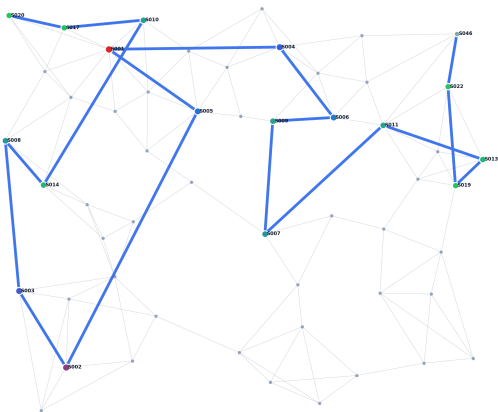
Post-optimization: local search swaps to improve an existing solution.

Heuristic	Length (km)	Stations	Ridership	Time (s)
$L_{\max} = 60$ km				
H1 (Extension)	59.8	18	6.22e12	0.05
H2-A (Insertion)	59.9	31	1.37e13	0.79
H2-A + post-opt	59.9	31	1.37e13	0.02
H2-B (Insertion)	59.9	36	9.57e12	1.04
H2-C (Insertion)	59.6	33	1.36e13	0.93

- H2 variants select **more stations** (31–36 vs. 18) within the same budget.
- H2-A achieves $\sim 2.2\times$ the ridership of H1.
- Post-optimization provides marginal improvement on this dataset.

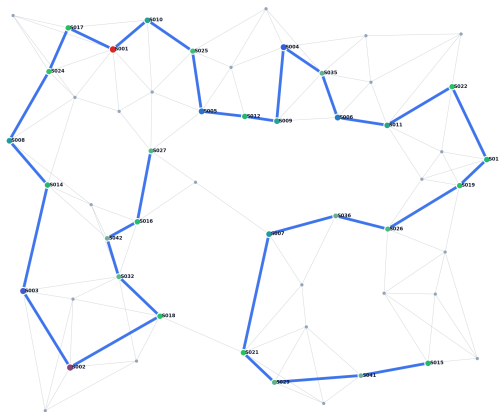
H1: Greedy Extension

H1_greedy_extension: 59.8 km, 6.2e+12 ridership, 18 stns



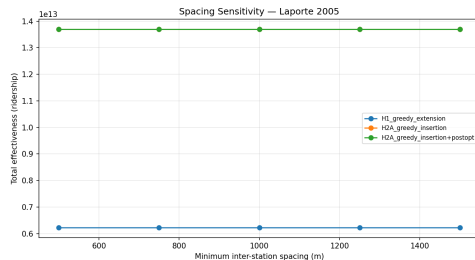
H2-A: Greedy Insertion

H2A_greedy_insertion: 59.9 km, 1.4e+13 ridership, 31 stns



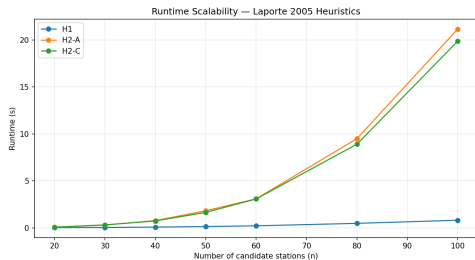
Laporte 2005 – Spacing Sensitivity

- Varying min inter-station spacing: 500m – 1500m
- Paper claims: H1 dominates for spacing $\geq 1250\text{m}$
- On Shanghai data: **H2-A consistently outperforms H1** across all spacing values
- Denser candidate set (54 stations, avg. $\sim 1\text{km}$) means H2's insertion flexibility helps even at tight spacing



Laporte 2005 – Runtime Scalability

- Sweep $n \in \{20, 30, 40, 50, 60, 80, 100\}$ stations
- H1: $\sim 0.8\text{s}$ at $n = 100$
- H2-A/H2-C: $\sim 20\text{s}$ at $n = 100$
- Confirms polynomial scaling
- H1 is $\sim 25\times$ faster than H2 at $n = 100$



Multi-line rail transit network design:

- Simultaneously select stations **and** line network
- Minimize total system cost:

$$\min \quad \phi_p \cdot C_{\text{passenger}} + \phi_o \cdot C_{\text{operator}} + \phi_c \cdot C_{\text{community}}$$

- $C_{\text{passenger}}$: travel time savings vs. car/bus
- C_{operator} : operating cost savings from mode shift
- $C_{\text{community}}$: construction cost (alignment + stations)

Key Insight

Simultaneous optimization of all lines $\approx$ **70% better** than optimizing lines individually.

Stage 1: Feasibility Screening

- GIS-based spatial filtering
- Our adaptation: reuse the 54-station pipeline (population, jobs, transport, commercial layers)
- Produces candidate station pool

Stage 2: Genetic Algorithm

- **Chromosome:** sequence of station indices per line
- **Fixed terminals:** selected from high-OD station pairs
- **Operators:**
 - Tournament selection (size 2)
 - Uniform crossover ($P_c = 0.7$)
 - Station-level mutation ($P_m = 0.3$)
- **Feasibility repair** after crossover/mutation

Chromosome Encoding

[Line_1_genes, Line_2_genes, ...]

Line_k = [term_a, stations..., term_b]

Each gene = station index from candidate pool.

Constraints

- 4–10 stations per line
- Min spacing: 1000m
- Max spacing: 3000m
- Max overlap: 30%

GA Parameters

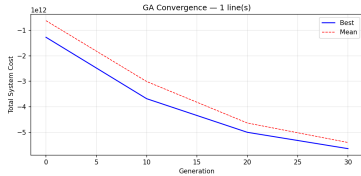
Parameter	Value
Population size	500
Generations	50
Crossover rate P_c	0.7
Mutation rate P_m	0.3
Tournament size	2

Fitness

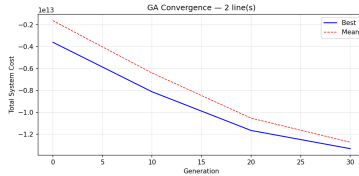
$$\min \phi_p C_p + \phi_o C_o + \phi_c C_c$$

Negative cost = savings (good).

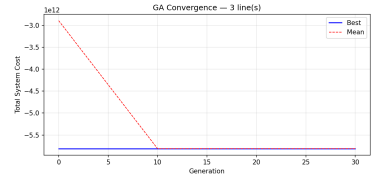
1-line



2-line



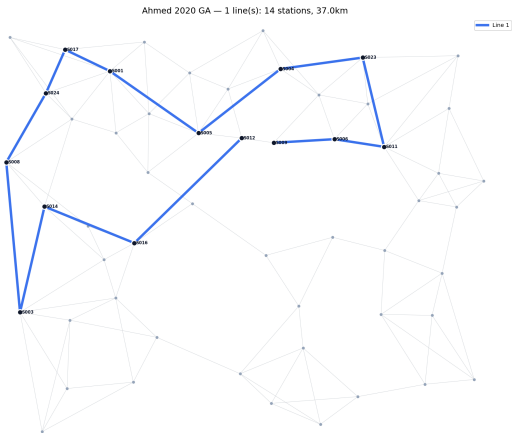
3-line



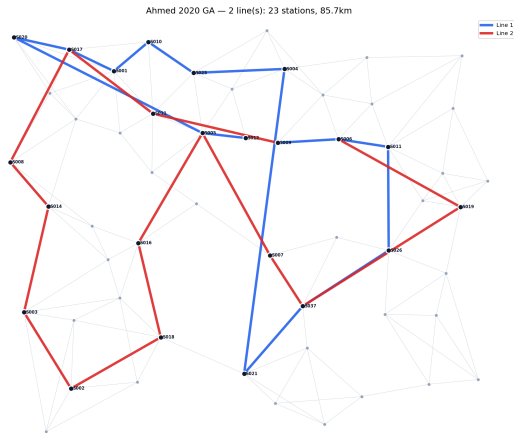
- Convergence within 20–30 generations for all scenarios.
- 2-line achieves lowest total cost (-1.33×10^{13}).

Ahmed 2020 – Network Visualizations

1-line network (14 stations, 37.0 km)



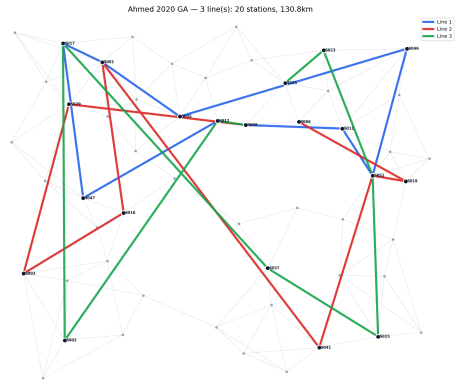
2-line network (23 stations, 85.7 km)



	Simultaneous	Individual
Total cost	-1.33×10^{13}	-1.13×10^{13}
Stations	23	28
Length (km)	85.7	80.1
Runtime (s)	58.3	68.9

18% improvement from simultaneous optimization.
(Paper reports 70% – difference due to simplified cost model and smaller candidate set.)

3-line network (20 stations, 130.8 km)



All Methods Comparison

Method	Stations	Length	OD Coverage	Lines	Runtime
Laporte H1	18	59.8 km	27.9%	1	0.05s
Laporte H2-A	31	59.9 km	54.0%	1	0.77s
Laporte H2-C	33	59.6 km	56.8%	1	0.88s
Ahmed 1-line	14	37.0 km	29.8%	1	33.5s
Ahmed 2-line	23	85.7 km	48.9%	2	56.3s
Ahmed 3-line	20	130.8 km	31.8%	3	58.1s
Greedy-OD	23	57.1 km	56.5%	3	0.01s
Greedy-BCR	10	15.7 km	27.3%	3	0.01s
GA (line-pool)	32	56.0 km	40.4%	3	0.90s

Key Findings

- ① **Laporte H2-A/H2-C achieve highest OD coverage** (54–57%) among all single-line methods.
- ② **H1 is $25\times$ faster** than H2 variants at $n = 100$, but selects fewer stations.
- ③ **Ahmed's multi-line GA** naturally handles transfer stations and network topology.
- ④ **Simultaneous optimization** outperforms individual line design by 18%.
- ⑤ **Greedy-OD baseline** is competitive with GA methods at negligible runtime.

Practical Implication

For rapid feasibility studies, greedy heuristics suffice. For detailed network design with transfers and multi-line coordination, GA-based methods provide meaningful improvements.

Complexity Comparison

Algorithm	Per-Iteration	Total Worst-Case
H1 (Laporte)	$O(n \cdot E ^2)$	$O(n^2 \cdot E ^2)$
H2-A/B/C (Laporte)	$O(n \cdot E ^3)$	$O(n^2 \cdot E ^3)$
Post-optimization	$O(E ^2 \cdot n)$ per pass	$O(E ^3 \cdot n)$
Ahmed GA (per gen)	$O(L \cdot N_s^2 \log N_s)$	$O(G \cdot P \cdot L \cdot N_s^2 \log N_s)$

- n : stations, $|E|$: edges, L : lines, N_s : stations per line
- P : population, G : generations
- All heuristics are **polynomial-time** despite NP-hard problem.

- Both papers' algorithms were **successfully reproduced** on a Shanghai 54-station dataset.
- **Laporte heuristics**: H2 insertion outperforms H1 extension in coverage; H1 wins on speed.
- **Ahmed GA**: simultaneous multi-line optimization is superior to individual line design.
- **Greedy baselines** remain surprisingly competitive for single-objective scenarios.
- Deviations from original papers are well-motivated and do not invalidate core algorithmic contributions.

Simplifications and Limitations

Simplifications Made

- Gravity-model OD instead of census-tract OD
- Pure coverage ($g_{ij} = 1$) instead of logit mode choice
- Simplified 3-component cost (vs. 25-equation model)
- 54 stations (vs. 4774 grid cells)

Limitations

- Shanghai data may not generalize to all cities
- Simplified cost model may underestimate community costs
- No real-world validation of ridership predictions
- GA convergence depends on parameter tuning

- **Sensitivity analysis** on Ahmed GA parameters (P_c , P_m , population size)
- **Multi-objective optimization**: Pareto front of coverage vs. cost
- **Integration with real transit data**: validate against Shanghai Metro ridership
- **Scalability testing** on larger networks (200+ stations)
- **Hybrid approaches**: use Laporte heuristics to seed the GA population

Thank you!

Questions?

Code: Rail-Line-Planning/line_network_model/

Report: doc/report/CS240_Report.tex